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Morphological studies of the olfactory organ of *Labeo gonius* (Ham.) from waterbodies of Gaya

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Abstract- Morphological studies of the olfactory organ in freshwater fishes' focus on understanding the structure and organization of this sensory system. These studies provide insights into how fishes detect and process olfactory stimuli in their environment. The current study of the olfactory organ of *Labeo gonius* (Ham.) from different waterbodies of Gaya used a biostatistical approach to determine the relationship between the fishes' lamellae and length. Additionally, a graph has been plotted for the current study. The ethmoid bone fossa of the skull, trapping the smell organs. The principal housing component is the lateral ethmoid. They are protected posterodorsally by the frontal bone, anterodorsally by the nasal bone, and ventrolaterally by the lachrymal bone. The olfactory rosette is attached to an auxiliary sac. The olfactory rosette is oval in shape and has multiple closely spaced lamellae. A single lamella is crescentic in shape and has a linguiform process. The radial lamellae, which vary in number from approximately 19 to 55, are arranged in transverse rows on both sides of the raphe. As the fish becomes longer, the number increases accordingly.

Key words: Olfactory organ, *Labeo gonius*, auxiliary sac, lamellae

INTRODUCTION

The *Labeo gonius* is a species of fish in the carp family, Cyprinidae. It is native to Pakistan, India, Bangladesh, and Burma, and it is known from Afghanistan and Nepal. The fish can grow to a length of 1.5 meters and a weight of 1.4 kg. This freshwater fish is typically found in rivers, though it can also occasionally be spotted in rice fields. In July and August, during the monsoon, it spawns. Aquaculture is used to raise this fish. Although it prefers to compete with some species, it can be raised alongside some other carps. The species is not often seen to be threatened because it is common and ubiquitous in some places. Overexploitation and habitat deterioration are two possible dangers to it. Fish olfactory organs are uniquely localized and have significant biological implications since they are involved in behavior such as food detection, sex recognition, defense against predators, and orientation and behavior of the parents.[1]

The environment provides physical and chemical stimulation to the sense organs. Skin receptors sense physical changes in heat flow or touch, whereas olfactory organs typically perceive chemical stimuli.[2] Visual stimuli entail changes in light intensity and are received through the inner ear or lateral line. Numerous researchers have examined the morphology and anatomy of the olfactory organs in teleostean fishes; among these, the following bear particular note. The accessory nasal sacs were described by Owen and Milne-Edwards (1874)[3] in relation

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to Mackerel and Anarrhichas' olfactory chamber. The coarse and fine anatomy of Sophora impar and Lophius piscatorius's olfactory organs are described in detail by Pereyaslawzeff (1876)[4]. The tubular kind of anterior nose in some fish, such as Conger and Anguilla, was described by Bateson (1889)[5]. Sticklebacks are known to have a single nostril, according to Solger (1894)[6]. In nearly every fish under study, the olfactory organ is located in a hollow depression in the ethmoid bone of the skull. The lateral ethmoid is the primary housing part. The nasal, frontal, and lachrymal tissues provide partial anterodorsal, posterodorsal, and ventrolateral protection, respectively. The olfactory rosette can be extended and compressed laterally, or it can be round or oval in shape.

MATERIALS & METHODS

The live specimens of *Labeo gonius* from various waterbodies in Gaya as well as the Phalgu River (Gaya) were gathered. Additionally, fish from Kakolat (near Nawada) were obtained. Both fresh and preserved specimens were dissected in order to examine the interaction between the brain and the skull bones and the olfactory organs. We removed the rosette and used a stereoscopic binocular microscope to inspect it. Through the binoculars, the arrangement of the olfactory lamellae was examined. Individual lamellae were also prepared as whole mounts.

The olfactory organs and became trapped in the skull's ethmoid bone fossa. The lateral ethmoid is the primary housing part. The frontal bone protects them posterodorsally, the nasal bone protects them anterodorsally, and the lachrymal bone protects them ventrolaterally. An auxiliary sac is fastened to the olfactory rosette. Much of the accessory sac's dorsal surface is protected from the outside by the mesethmoid's lateral wings and lachrymal. The enlarged anterior end of the palatine serves as the accessory sac's ventral resting place.

The anatomical link between the brain and the olfactory rosette is demonstrated by dissecting the head from the dorsal side. The olfactory bulb is a structure that resembles a vase, with an exterior, anterior neck-like portion and a mesial, posterior rounded body. The latter receive olfactory fibers from the matching rosette and abut the mesial surface of the olfactory chamber at the level of the olfactory foremen in the lateral ethmoid. The long, paired olfactory tracts are present. Each fiber emerges from the olfactory bulb and passes through the frontal portion of the skull, which is held ventrally by the orbitosphenoid, before turning back and inward. It finally comes to an end along its anteroventral aspect in the telencephalon, where it forms an approximate 23° angle with the body's anteroposterior axis.

The olfactory rosette has several closely spaced lamellae and is oval in form. A solitary lamella has a linguiform process and a crescentic shape. On both sides of the raphe, the radial lamellae, which range in number from roughly 19 to 55, are organized in transverse rows. The number rises in proportion to the fish's increasing length.

RESULT & DISCUSSION

After taking into account the aforementioned findings, it is clear that freshwater fishes have paired olfactory organs located in the form of an olfactory pit on the dorsal portion of the head. The front inlet and the posterior outlet are the two distinct openings found in each pit. Some fish have two separate regions within their olfactory chamber: the actual olfactory chamber, which includes the olfactory rosette, and the accessory nasal sac. The olfactory organ is located in a hollow depression in the ethmoid bone of the skull and is virtually always set in place relative to the other skull bones in fish. Small changes can be found in the various degrees of the olfactory organ's features, particularly in relation to the rosette and olfactory lamellae.[7]

In order for the inflowing water to directly contact the olfactory lamellae, the anterior inlet opens up immediately above the olfactory resette. *Labeo gonius* has a posterior exit that is crescentic in shape. The olfactory chamber of *Labeo gonius* is partially divided into two recesses by the graved skin that extends into the chamber from between the two openings. This structure resembles a curtain and hangs over the rosette. Water motions in the olfactory chamber are facilitated and controlled by this structure, which resembles a curtain. Burne (1909)[8] made extremely similar observations about certain fish in the clupeidae family. It is discovered that there is a single asscessory nasal sac located ventrally to the olfactory rosette, or the lachymal sac in *Labeo gonius*. Burne has made this remark (1909)[8]. The opening and closing of the mouth for breathing directly causes the dilatation and compression of the sacs. It assists in keeping the anterior inlet's water flow across the olfactory lamellae mild as well as in preventing debris buildup in the olfactory chamber.

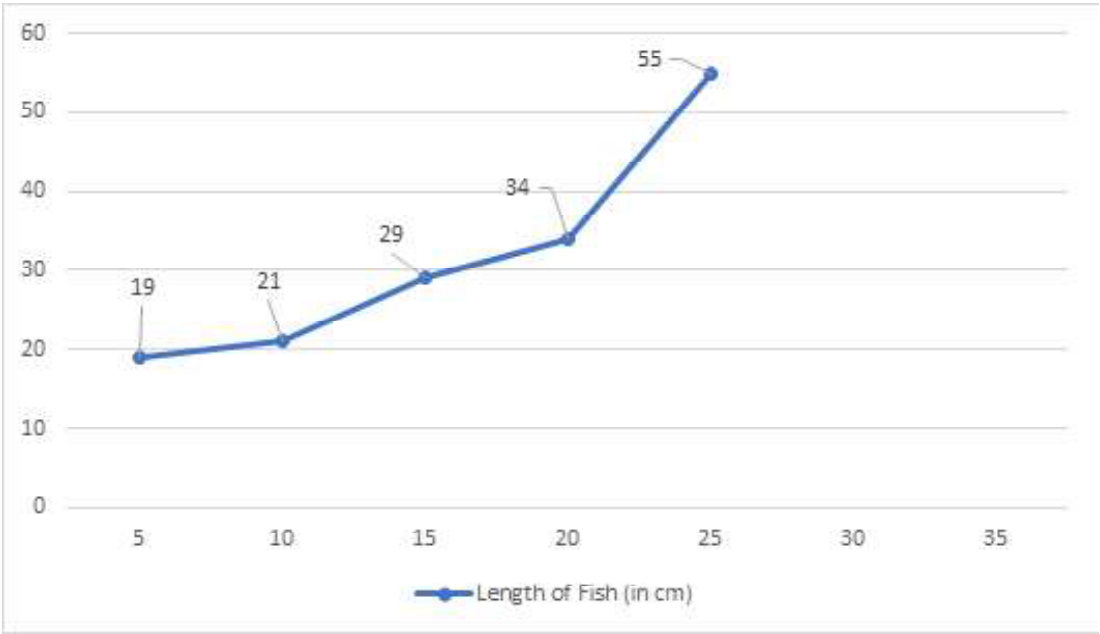


Fig. 1- Samples of *Labeo gonius* collected from different waterbodies of Gaya

Table 1:- Showing the number of olfactory lamelle increases with the corresponding increase in the length of the fish

Sl.No.	T.L. of fish mm	S.L. of fish mm	No. of olfactory lamelle		Total no. of lamella in both rosettes
			Left rosette	Right rosette	
1	80	65	9	10	19
2	90	70	10	11	21
3	120	100	15	14	29
4	1 35	115	17	17	34
5	200	180	28	27	55

Graph 1:- Showing the correlation between the length of a fish and the number of lamellae in the olfactory organ



CONCLUSION

Morphological studies of the olfactory organ in freshwater fish *Labeo gonius* provide a comprehensive view of how these animals detect and process odors. By employing a range of microscopic and molecular techniques, researchers can gain insights into the structural adaptations of the olfactory system, compare different species, and understand the functional implications of olfactory morphology. This knowledge is crucial for ecological research, environmental monitoring, and applications in aquaculture and conservation.

A comprehensive study of the olfactory organ in freshwater fishes involves examining the anatomical structure, cellular composition, and functional aspects of the olfactory system. Histological and histochemical techniques, combined with behavioral and environmental analyses, provide a detailed understanding of how these fishes detect and process odors. This knowledge is valuable for ecological research, environmental monitoring, species conservation, and aquaculture practices.

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